Data Interpretation Exercise-CO1-AT1

Question 1: Mapping Data onto Aesthetics

Dataset

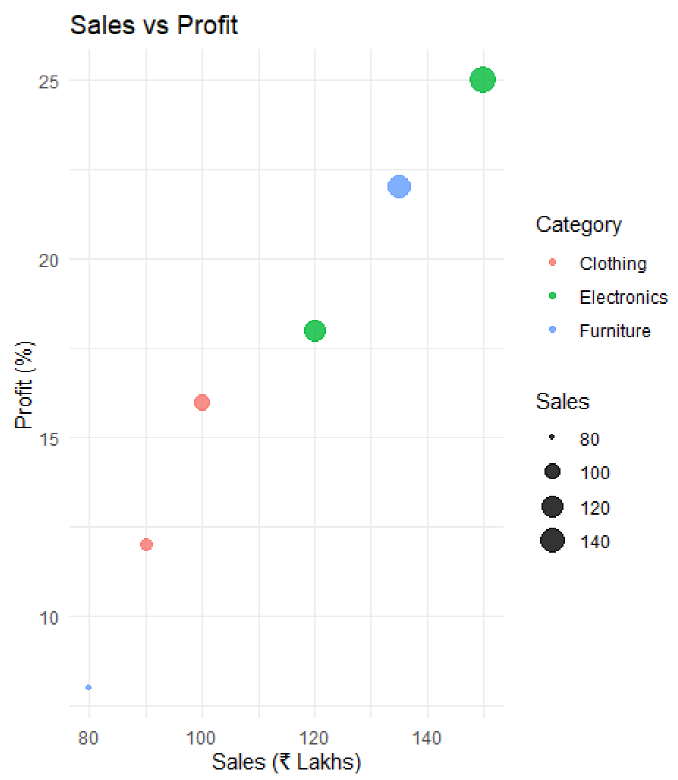
R

```r
library(ggplot2)


data <- data.frame(
  Product = c("P1","P2","P3","P4","P5","P6"),
  Sales = c(120,90,150,80,135,100),
  Profit = c(18,12,25,8,22,16),
  Category =
c("Electronics","Clothing","Electronics","Furniture","Furniture","Clothing")
)
```

1. R Program (Scatter Plot)

R

```r
library(ggplot2)

ggplot(data, aes(x = Sales,
             y = Profit,
             color = Category,
             size = Sales)) +
  geom_point(alpha = 0.8) +
  labs(title = "Sales vs Profit",
     x = "Sales (₹ Lakhs)",
     y = "Profit (%)") +
  theme_minimal()
```



2. Aesthetic Mapping

X-axis → Sales

Y-axis → Profit

Color → Category

Size → Sales

3. Interpretation

Higher sales generally correspond to higher profit.

Electronics products have the highest sales and profit.

Furniture and Clothing have comparatively lower values.

4. Type of Scale

Variable	Scale Type
Sales	Continuous
Profit	Continuous
Category	Categorical (Discrete)

5. Alternative Visualization

Bubble Chart (or grouped bar chart) because it can display sales, profit, and category simultaneously.

Question 2: Coordinate Systems

Dataset

R
```
library(ggplot2)

direction <- data.frame(
  Direction = c("N","NE","E","SE","S","SW","W","NW"),
  Frequency = c(18,12,20,15,22,11,16,14)
)
```

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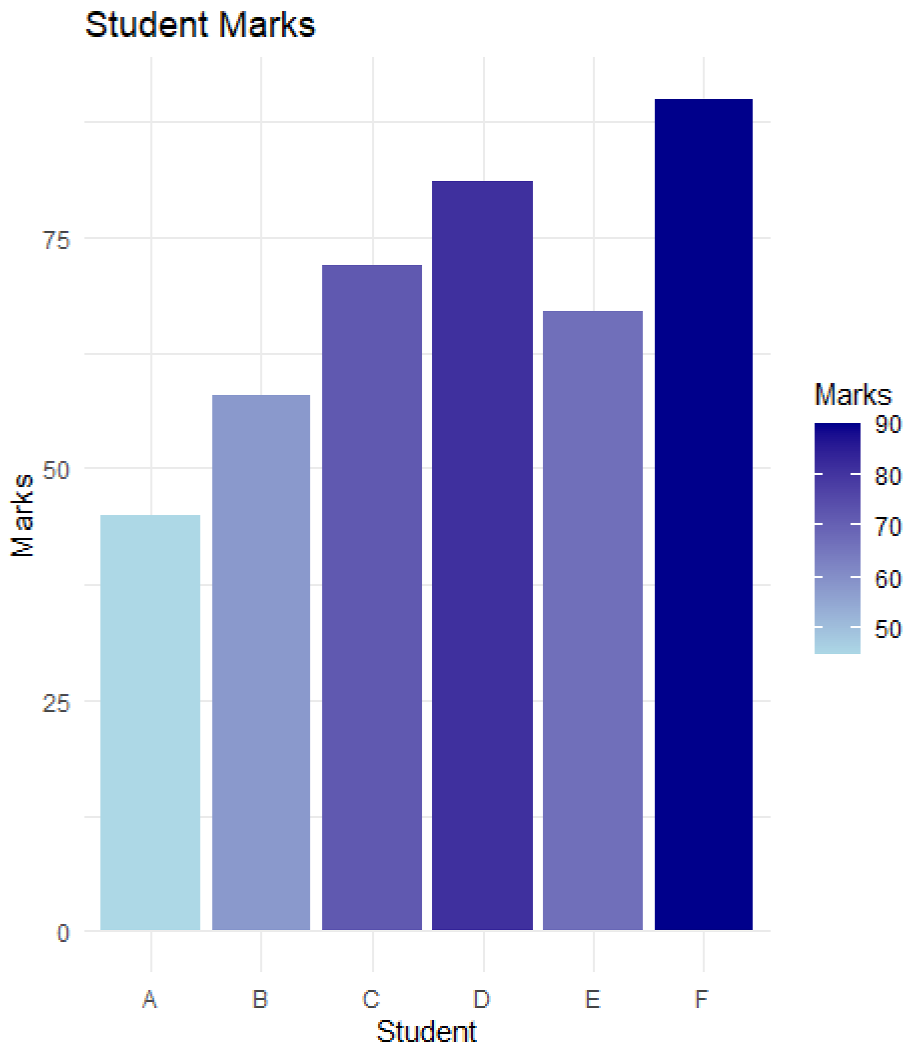
1. Cartesian Coordinate System

R
```
ggplot(direction,
       aes(x = Direction,
           y = Frequency)) +
  geom_bar(stat = "identity",
           fill = "steelblue") +
  labs(title = "Direction Frequency (Cartesian)",
       x = "Direction",
       y = "Frequency") +
  theme_minimal()
```



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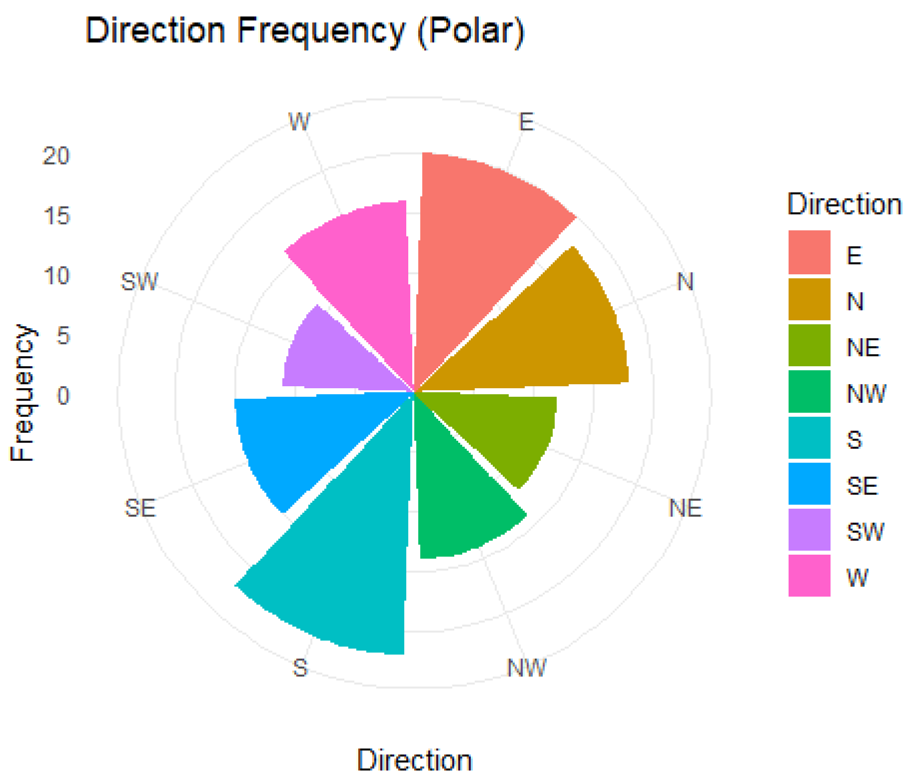
2. Polar Coordinate System

R

```
ggplot(direction,
     aes(x = Direction,
        y = Frequency,
        fill = Direction)) +
 geom_bar(stat = "identity") +
 coord_polar() +
 labs(title = "Direction Frequency (Polar)") +
 theme_minimal()
```



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3. Comparison

Cartesian

Easier to compare frequencies.

More accurate for reading values.

Polar

Better represents directional data.

Attractive and intuitive.

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4. More Appropriate Coordinate System

Polar Coordinate System

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5. Justification

Directional data naturally represents angles (N, NE, E...), making the polar coordinate system more meaningful.

Question 3: Color Scales

Dataset

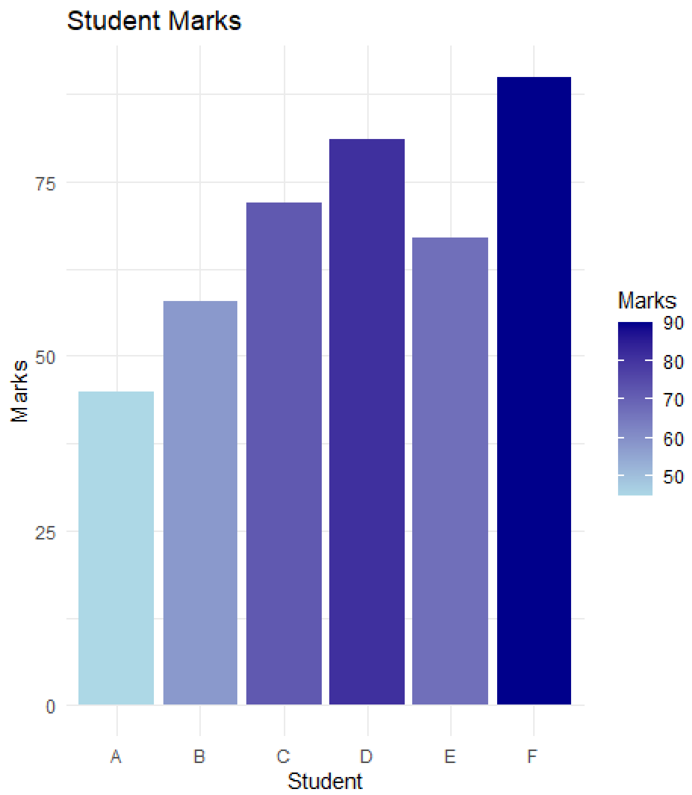
```R
R
library(ggplot2)

marks <- data.frame(
  Student = c("A","B","C","D","E","F"),
  Marks = c(45,58,72,81,67,90)
)
```

---

1. R Program

```R
R
ggplot(marks,
     aes(x = Student,
        y = Marks,
        fill = Marks)) +
  geom_bar(stat = "identity") +
  scale_fill_gradient(low = "lightblue",
                high = "darkblue") +
  labs(title = "Student Marks",
     x = "Student",
     y = "Marks") +
  theme_minimal()
```



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2. Highest and Lowest Marks

Highest → **F (90)**

Lowest → **A (45)**

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3. Why Sequential Color Scale?

Marks are **continuous numerical values**, so gradual color intensity effectively represents increasing values.

4. Color-blind Friendly Palette

Use the **Viridis** palette.

R

```
scale_fill_viridis_c()
```

5. Sequential vs Categorical Color Scale

Sequential	Categorical
Used for continuous data	Used for discrete groups
Color intensity changes gradually	Different colors represent categories
Example: Marks	Example: Region

Question 4: Highlighting with Color

Dataset

R

```
library(ggplot2)

revenue <- data.frame(
  Region = c("North","South","East","West","Central"),
  Revenue = c(820,760,790,1010,770)
)

revenue$Highlight <- ifelse(revenue$Revenue ==
                        max(revenue$Revenue),
                      "Highest",
                      "Others")
```

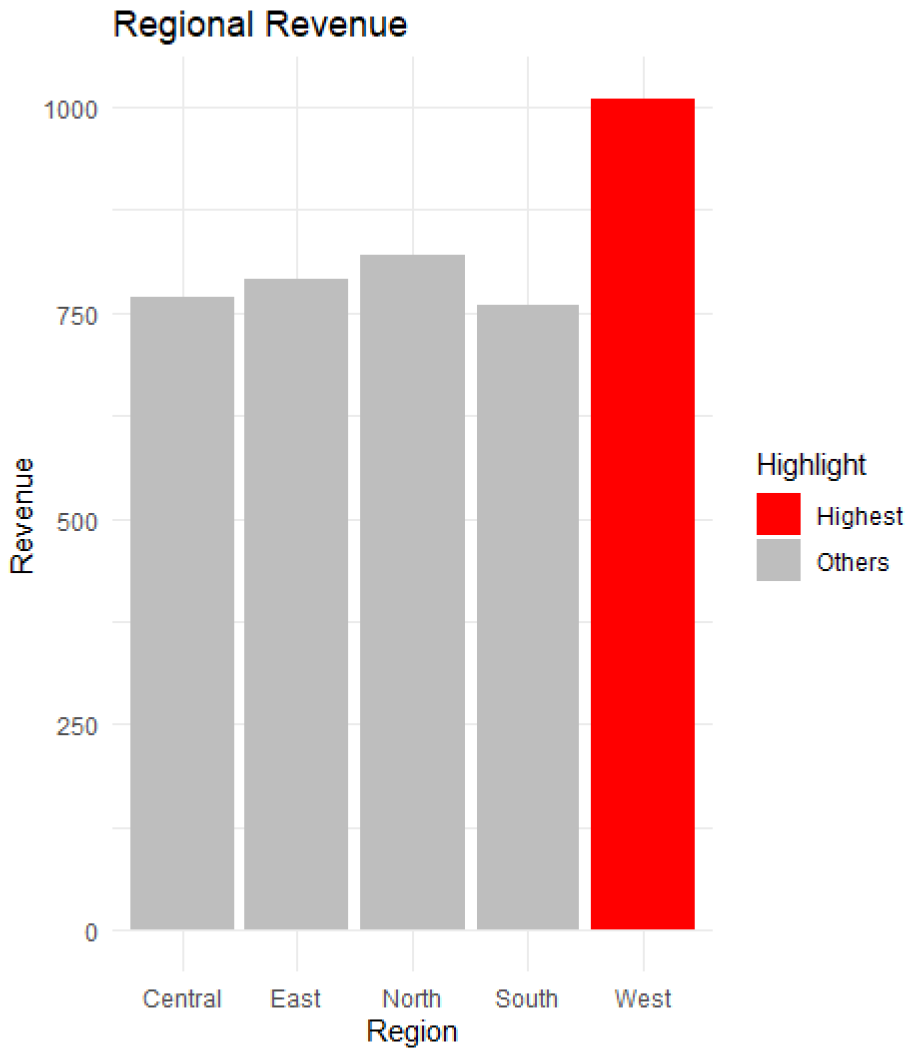
---

1. R Program

R

```
ggplot(revenue,
      aes(x = Region,
          y = Revenue,
          fill = Highlight)) +
  geom_bar(stat = "identity") +
  scale_fill_manual(values =
                c("Highest"="red",
                  "Others"="gray")) +
  labs(title = "Regional Revenue") +
```

theme_minimal()



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2. How Highlighting Helps

Highlighting immediately draws attention to the most important data point.

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3. Highest Performing Region

West (₹1010 Crores)

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4. Effect of Excessive Colors

Makes charts confusing.

Reduces readability.

Distracts viewers.

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5. Alternative Highlighting Technique

Add labels/annotations to the highest bar.

Increase bar border thickness.

Use different bar patterns.

Question 5: Integrated Visualization

Dataset

R
```
library(ggplot2)

city <- data.frame(
  City = c("A","B","C","D","E"),
  Population = c(8.2,5.6,9.4,6.8,4.9),
  AQI = c(110,180,90,240,150),
  Rainfall = c(850,430,1200,520,680),
  Region = c("North","South","North","East","West")
)
```

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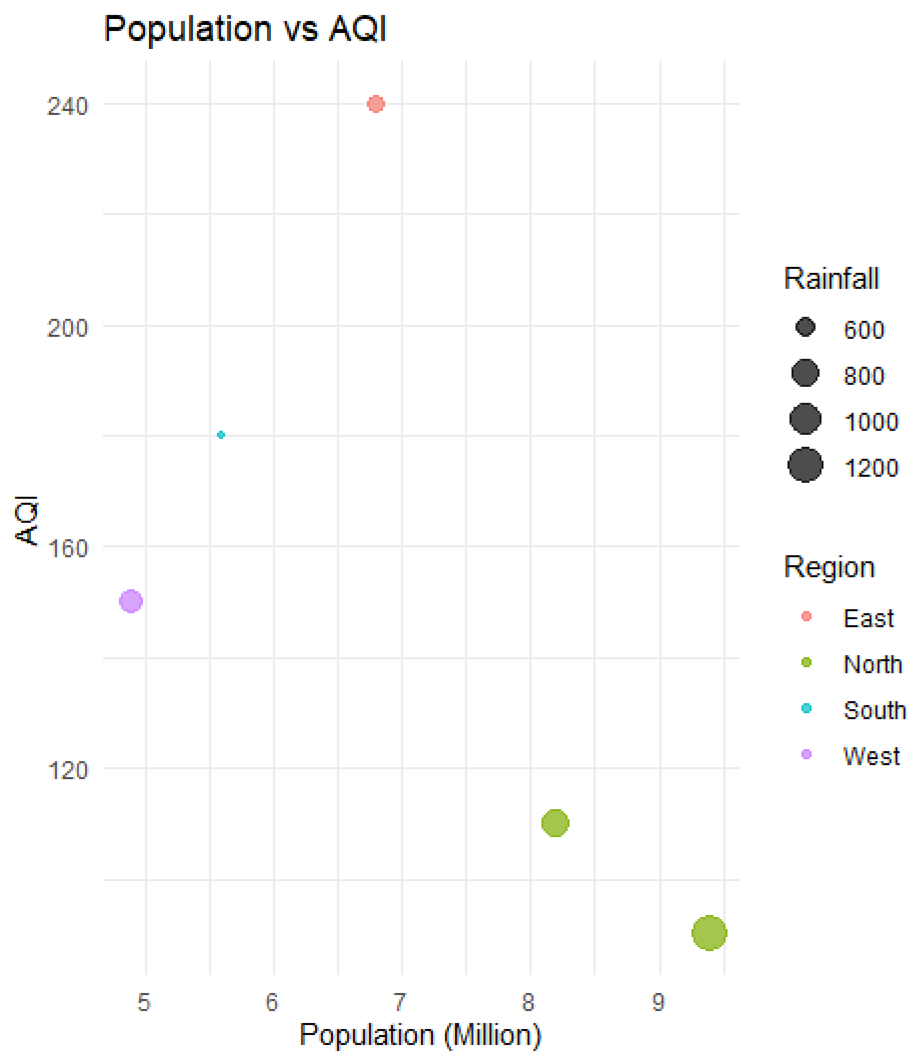
1. R Program (Bubble Chart)

R
```
ggplot(city,
       aes(x = Population,
           y = AQI,
           size = Rainfall,
           color = Region)) +
  geom_point(alpha = 0.7) +
  labs(title = "Population vs AQI",
       x = "Population (Million)",
       y = "AQI") +
  theme_minimal()
```



2. Aesthetic Mapping

Variable	Aesthetic
Population	X-axis
AQI	Y-axis
Rainfall	Bubble Size
Region	Color

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3. Interpretation

There is no strong linear relationship between Population and AQI.

City D has the highest AQI despite having only a moderate population.

City C has the largest population but a relatively low AQI.

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4. Continuous and Categorical Variables

Continuous Variables

Population

AQI

Rainfall

Categorical Variable

Region

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5. Improvement

Add city labels to each bubble for easier identification.

R

```
geom_text(aes(label = City), vjust = -1)
```

This makes the chart easier to interpret without referring back to the dataset.